DESIGNING AUTHENTIC INDUSTRY-ENGAGED ASSESSMENT FOR PROFESSIONAL COMPETENCE IN BUSINESS INTELLIGENCE

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ABSTRACT

Engineering programs increasingly seek to align student learning with professional practice, yet meaningful industry collaboration often exposes misalignment between academic assessment norms and workplace expectations. This paper presents a CDIO Implementation case describing the redesign of an undergraduate business intelligence course to strengthen industrial relevance while supporting student professional development. The redesign integrated semester-long, industry-inspired project work structured around team-based learning and competence-oriented assessment, emphasizing authentic data, reproducible workflows, and professional collaboration tools. A single industrial partner contributed real-world datasets, problem framing, and feedback, enabling students to engage with constraints and quality standards characteristic of professional practice. Assessment and feedback were redesigned to balance team performance with individual contribution through visible artifacts (e.g. version control activity) and structured reflection, addressing persistent challenges related to workload, accountability, and fairness. Evidence from iterative course evaluations, student reflections, graduate feedback, and industry partner input indicates increased student engagement, improved coherence across learning activities, and clearer alignment between assessment criteria and professional expectations. At the same time, the collaboration revealed tensions between academic tolerance for developmental errors and industry demands for reliability and trustworthiness, highlighting the importance of explicitly preparing students to communicate differently with academic and professional audiences. The paper reflects on these tensions and outlines design principles for industry-engaged CDIO courses, including expectation alignment early in the course, sustained use of a shared industrial context, and assessment practices that foreground professional standards without sacrificing formative learning space. The case offers transferable insights for CDIO programs seeking to leverage industrial collaboration as a catalyst for student professional development through innovative assessment and feedback.

KEYWORDS

Industry collaboration, professional competence, assessment and feedback, team-based learning, CDIO Standards: 5, 7, 8, 11

INTRODUCTION AND MOTIVATION

Engineering education increasingly emphasizes alignment between academic learning and professional practice. Frameworks such as CDIO foreground not only disciplinary knowledge, but also the personal and interpersonal competencies required for graduates to function effectively in complex, real-world environments (Malmqvist et al., 2022). In parallel, industry increasingly expects graduates to collaborate effectively, work with authentic data, and produce credible, reproducible results.

Despite broad agreement on these goals, tensions persist between academic assessment practices and professional standards. In higher education, assessment often supports learning as a developmental process in which errors are tolerated and encouraged. Evaluation in industry settings places substantial emphasis on understanding the data, its provenance, limitations, and business implications, often more so than on demonstrating detailed knowledge of specific algorithms. These differences become especially visible in courses involving industry collaboration, where academic norms and workplace expectations intersect.

Business Intelligence provides a revealing context in which to examine this tension. As a discipline integrating data analysis, information technology, and organizational decision support, it requires practitioners to combine technical competence with professional judgment and clear communication of assumptions, limitations, and implications.

The aim of this paper is to describe and analyze this course redesign, with particular focus on how assessment and feedback function as mechanisms for professional competence development in an industry-engaged project context, and to reflect on resulting tensions, trade-offs, and transferable design principles.

COURSE AND INDUSTRY CONTEXT

Business Intelligence is a 6 ECTS elective aimed at advanced undergraduate and early graduate students in Industrial Engineering and related data-intensive programs. The course runs over one semester, organized into five thematic two-week cycles followed by a capstone project that consolidates prior work within a sustained industrial context (Fig. 1).

Industry engagement is provided by a single partner organization contributing authentic data and domain framing. The same partner participated in two consecutive annual course offerings, enabling informed comparison between iterations of the course design and providing an external perspective on changes in coherence, credibility, and professional relevance of student outputs. Students may augment the context with publicly available sources, but all analyses and deliverables remain anchored in the shared organizational setting.

DESIGN FRAMEWORK AND ALIGNMENT WITH CDIO

The course redesign intentionally aligns professional competence goals, learning activities, and assessment practices within a CDIO-inspired framework. Assessment and feedback were designed as mechanisms shaping development toward professional standards rather than as terminal verification of outcomes.

The framework combines team-based learning, problem- and experience-oriented pedagogy, and competence-oriented assessment within a sustained industrial context, consistent with CDIO Standards addressing design–implement experiences, integrated learning, active learning, and learning assessment (Malmqvist et al., 2020). While prior CDIO work has highlighted challenges in assessing interpersonal competencies in project courses and proposed mechanisms such as peer assessment and logbooks (Hansen and Sindre, 2023; Huet et al., 2008; Junaid et al., 2018), there has been comparatively less attention to their interaction with industry-facing expectations of credibility and trust in sustained collaborations.

Professional Competence in a CDIO Perspective

Professional competence is understood as the integrated ability to analyze open, data-rich problems, locate and critically evaluate information, and collaborate effectively under realistic constraints. This aligns with the CDIO Syllabus 3.0 emphasis on personal, interpersonal, and system-building competencies alongside disciplinary knowledge (Malmqvist et al., 2022).

To operationalize these goals, the redesign drew on the LOUIS competence framework developed within the Aurora Universities Network (2022). From the sixteen LOUIS competence domains, three were selected as design anchors: (i) problem solving, (ii) information literacy, and (iii) teamwork. These domains were chosen for their centrality to Business Intelligence practice and suitability for competence-oriented assessment in an industry-engaged course.

Focusing on a limited set of domains enabled constructive alignment across learning outcomes, activities, and assessment while maintaining coherence in the presence of authentic project complexity (Biggs and Tang, 2015).

Learning, Assessment, and Feedback Design Principles

The redesign rests on three complementary principles.

Team-Based Learning as structure: Team-Based Learning (TBL) provided the organizational backbone, structuring each cycle into preparation, team readiness, team application, and individual reflection (Fink, 2013; Michaelsen et al., 2004; Sibley and Ostafichuk, 2014). The team application phase (tAPP) was central, requiring collaborative analysis of authentic, industry-inspired data.

Problem- and experience-oriented learning: Learning activities were framed as open-ended, data-intensive problems within a sustained industrial context, drawing on problem-based and experiential learning principles (Barrows, Tamblyn, et al., 1980; Bruce and Bloch, 2012; Kolb, 2014). This continuity enabled progressive refinement of analytical approaches rather than repeated onboarding to unrelated datasets.

Assessment-as-development: Assessment criteria prioritized professional standards such as reliability, reproducibility, and credibility, while preserving formative learning space, a challenge frequently noted in CDIO project-based assessment contexts (Hansen and Sindre, 2023). Visible artifacts (e.g. version-control histories, code reviews) and structured reflection supported traceability within team outputs. Rubrics aligned with the selected LOUIS domains were reused across cycles, enabling iterative improvement and making quality criteria explicit across tasks (Biggs and Tang, 2015; Wiggins and McTighe, 2005).

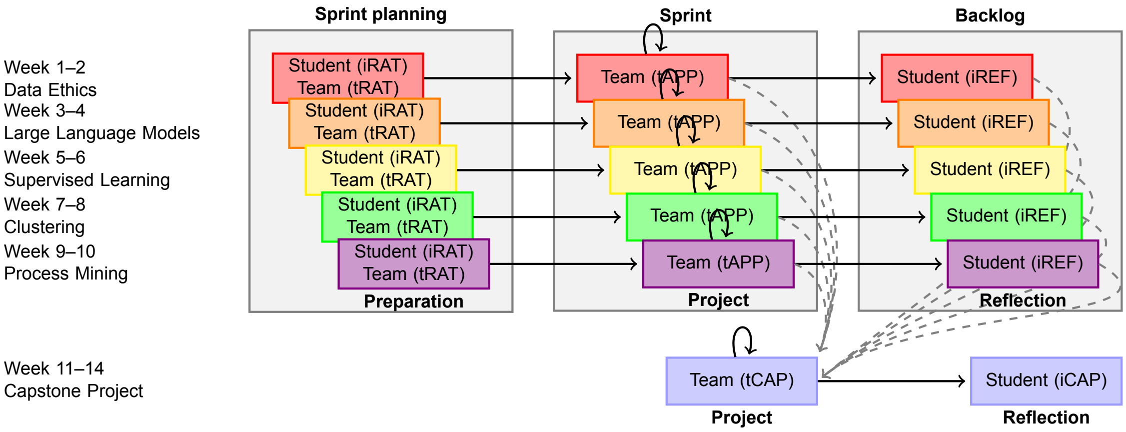

Figure 1. Course flow illustrating the structure of thematic team-based learning cycles and their integration into the capstone project. Each cycle includes preparation (iRAT/tRAT), team application (tAPP), and individual reflection (iREF). Reflections serve as forward-oriented notes and backlogs that inform the capstone project (tCAP), supporting continuity, iteration, and professional competence development across the semester.

Pedagogical Mechanisms and Tooling Choices

Preparatory materials and method overviews were assigned outside scheduled class time, reserving contact time for application, discussion, and feedback (flipped classroom; Tucker, 2012). Within each TBL cycle, individual readiness activities (iRAT) required students to propose context-specific uses of analytical methods, while team readiness discussions (tRAT) consolidated approaches prior to project execution.

GitHub Classroom was adopted as the primary collaboration environment, reflecting common professional practice in data-intensive and software-supported engineering work. The platform supported version-control, peer review, and reproducibility, enabling assessment to recognize both team outcomes and individual contributions through observable work traces.

Each two-week cycle was aligned with agile development practices (Schwaber and Beedle, 2001). Planning activities in tRAT functioned as lightweight sprint planning, tAPP as execution, and individual reflection (iREF) as a structured retrospective. Reflection was forward-oriented: students documented unresolved issues, methodological limitations, and improvement ideas that they revisited during the capstone project.

Course Flow and Iterative Integration

Figure 1 illustrates the overall course flow, showing how five thematic TBL cycles feed into a final capstone project. Each cycle comprises preparation (iRAT/tRAT), team project work (tAPP), and individual reflection (iREF). Dashed connections indicate how both team outputs and individual reflections contributed to the capstone project (tCAP). The capstone project functioned as a sustained integrative task in which teams revisited and consolidated insights, feedback, and unresolved issues from earlier cycles, enabling assessment and reflection to accumulate toward professional standards rather than reset between tasks (Kecskes and Kerrigan, 2009).

By explicitly carrying insights, constraints, and unresolved questions forward into the capstone, the course avoided treating assessment tasks as isolated events. Instead, assessment and feedback accumulated across cycles, reinforcing learning as an iterative, developmental process aligned with professional practice.

Alignment with CDIO Standards

The redesigned course aligns with CDIO Standards in several ways. Standard 5 (Design–Implement Experiences) is addressed through semester-long, industry-inspired project work that requires students to design analytical approaches and implement reproducible solutions. Standard 7 (Integrated Learning Experiences) is reflected in the intentional integration of professional competencies with business intelligence methods within a sustained industrial context. Standard 8 (Active Learning) is operationalized through TBL cycles, flipped preparation, and in-class application requiring reasoning, collaboration, and decision-making. Finally, Standard 11 (Learning Assessment) is addressed by competence-aligned rubrics, artifact-based accountability, and iterative feedback loops that balance formative learning with professional expectations. Together, these elements position assessment and feedback as central drivers of development toward professional standards in a CDIO-aligned business intelligence course.

COURSE REDESIGN: ASSESSMENT AND FEEDBACK ARCHITECTURE

This section describes how assessment and feedback were redesigned to support professional competence development rather than primarily serving as summative measurement. Building on the design framework outlined earlier in this paper, assessment was treated as a central element of the learning architecture, shaping student behavior, collaboration, and engagement with professional standards throughout the semester.

The redesign responded explicitly to tensions observed in earlier iterations of the course, including perceived misalignment between academic assessment practices and industry expectations, uneven workload distribution in team projects, and limited continuity between individual project cycles. Addressing these challenges required rethinking the role of assessment artifacts, feedback timing, and individual accountability within collective work.

The findings reported are based on a single course and one industry partner across two iterations, and draw on qualitative evidence (observations, reflections, and feedback). The paper therefore aims to provide analytically transferable insights rather than generalizable claims.

From Assessment as Measurement to Assessment as Development

We adopted a developmental view of assessment, in which criteria and feedback guided progression toward professional standards over time. The final summative rubric was introduced early and reused formatively across project cycles with scaled weight, allowing students to rehearse performance against final criteria while retaining space for revision. Early exploration and even errors were acceptable if recognized, documented, and addressed. Aligned rubrics–covering analytical soundness, transparency, reproducibility, and stakeholder relevance–were reused across cycles and the capstone so that feedback accumulated and informed later work rather than fragmenting across tasks. Version-control histories and pull-request records provided concrete evidence of whether and how teams responded to feedback between iterations.

Industry-Engaged Project Structure

A key element of the assessment architecture was the use of a shared, industry-inspired project context spanning the entire semester. A single industry partner provided an authentic dataset and participated in problem framing, milestone discussions, and final presentations. Students were free to augment the provided data with additional sources, but all work remained anchored in the same organizational and analytical context.

This design choice addressed two recurring challenges observed in earlier course iterations: repeated onboarding to unrelated datasets and weak coherence between project cycles. By maintaining a stable context, students could progressively deepen their understanding of the domain, refine analytical approaches, and revisit earlier assumptions as new methods were introduced.

Assessment requirements emphasized reproducibility and traceability. All analyses were developed in shared repositories, and deliverables consisted not only of reports and presentations, but also of executable code and documented workflows. This structure mirrored professional expectations in data-intensive work and reinforced importance of credible, transferable results.

Industry feedback emphasized trustworthiness and usability, making visible how even minor analytical flaws can erode confidence–an intentional contrast to academic tolerance for developmental error that students learned to navigate.

Balancing Team Performance and Individual Accountability

Team-based project work was central to the course, mirroring professional practice in business intelligence. However, team assessment introduces well-known challenges related to unequal contribution, free-riding, and perceived unfairness, as widely recognized in CDIO project courses (Huet et al., 2008). Rather than relying solely on peer grading or individual exams, the redesign adopted a dual assessment strategy.

Team deliverables were assessed collectively with respect to analytical quality, coherence, and professional relevance. In parallel, individual contribution was assessed through visible artifacts generated during project work, including version-control activity, code reviews, and documented design decisions. Structured individual reflection complemented these artifacts by requiring students to articulate their contributions, uncertainties, and learning needs.

This approach allowed individual accountability to be supported without fragmenting team outcomes. Students reported that this transparency clarified expectations and reduced uncertainty about how their individual efforts were recognized within the team context.

Feedback Loops and Iterative Learning

Feedback was integrated at multiple levels and stages of the course. Instructor feedback focused on methodological soundness, alignment with competence criteria, and progression across cycles. Peer feedback occurred primarily within teams, through code review and collaborative problem-solving, and across teams during selected review moments.

Individual reflection (iREF) functioned as a forward-oriented practice: students documented unresolved issues, limitations, and improvement opportunities that were revisited during the capstone. This is consistent with CDIO approaches to reflection as professional practice rather than retrospective reporting (Junaid et al., 2018). As illustrated in Figure 1, these reflections formed a personal backlog that students revisited during the capstone project, enabling feedback from earlier cycles to inform later design decisions.

Industry feedback was strategically positioned at key points, particularly during the capstone phase. While academic feedback often tolerated developmental imperfection, industry feedback prioritized trustworthiness, clarity, and practical usability. The coexistence of these feedback modes challenged students to reconcile academic and professional expectations and to adapt their work and communication accordingly.

Together, these feedback loops supported iterative learning across the semester and reinforced assessment as a developmental process, closely aligned with professional practice while remaining pedagogically supportive.

STUDENT AND INDUSTRY RESPONSES

Evidence of the impact of the redesigned assessment and feedback architecture was gathered from multiple sources, including course evaluations, structured student reflections, informal graduate feedback, and feedback from the participating industry partner across two consecutive course offerings.

Reported outcomes across course evaluations, reflections, graduate feedback, and partner input converged on three themes: (i) engagement and perceived relevance remained high without formal attendance requirements, reflected in consistently high attendance and active participation; (ii) early-career transfer was evident in students' use of reproducible workflows, collaborative problem-solving, and explicit documentation of assumptions and limitations; and (iii) partner feedback valued improved coherence and credibility of outputs while highlighting that small analytical flaws can undermine trust–an expectation difference we now make explicit to students.

TENSIONS, TRADE-OFFS, AND LESSONS LEARNED

Industry engagement exposed productive tensions central to the redesign. Most prominently, academic tolerance for developmental error contrasted with industry expectations of reliability and trustworthiness. Accepting developmental error proved pedagogically valuable, but it also complicated students' calibration of when exploratory imperfection was acceptable and when it conflicted with professional expectations of reliability.

Team-based assessment presented a second trade-off. Large collaborative projects supported professional skill development but heightened concerns about workload distribution and fairness. Visible artifacts and structured reflection reduced ambiguity around individual contribution, though increased transparency also amplified student sensitivity to perceived inequities.

Tooling introduced a further tension. Professional collaboration platforms enhanced authenticity and reproducibility but imposed both an initial learning burden and substantial instructor effort through ongoing review and feedback. This front-loaded investment, however, streamlined final assessment by making revisions and responses to feedback clearly visible across iterations. Relocating tool onboarding earlier in the curriculum emerged as a pragmatic adjustment to preserve realism while reducing cognitive load during complex project work.

Collectively, these lessons indicate that industry-engaged assessment requires deliberate negotiation between realism and pedagogical support. Designing explicitly for such tensions–rather than attempting to eliminate them–proved essential for maintaining both learning quality and professional credibility.

TRANSFERABLE DESIGN PRINCIPLES FOR CDIO PROGRAMS

1. **Anchor assessment in a sustained industrial context.** Maintaining a shared, semester-long industry context supports coherence across learning activities and enables cumulative competence development, reducing fragmentation between project cycles.
2. **Design assessment as a developmental process.** Align assessment criteria across formative and summative tasks so that feedback accumulates over time and guides students toward professional standards rather than functioning as isolated judgments.
3. **Make individual contribution visible within team work.** Combine team-level outputs with artifact-based indicators and structured reflection to support accountability and fairness without undermining collaborative learning.
4. **Use reflection as forward-oriented professional practice.** Frame reflection as documentation of unresolved issues and improvement opportunities that inform subsequent design decisions, rather than as retrospective self-evaluation alone.
5. **Explicitly manage academic–industry expectation gaps.** Surface differences between developmental learning tolerance and professional reliability requirements early, and prepare students to adapt quality standards and communication to audience and context.
6. **Introduce professional tooling with intentional scaffolding.** Select tools that support authenticity and reproducibility, but ensure early alignment and onboarding to prevent tool mastery from overshadowing learning objectives.

These principles emphasize assessment and feedback as central design elements in CDIO-aligned courses and offer practical guidance for leveraging industry collaboration to support professional competence development.

CONCLUSIONS AND FUTURE WORK

This paper presented a CDIO-aligned redesign of an industry-engaged Business Intelligence course in which assessment and feedback act as primary mechanisms shaping development toward professional practice. The approach is aligned with the CDIO Syllabus 3.0 and Standards 5, 7, 8, and 11, combining team-based learning, authentic project work, and competence-oriented assessment within a sustained industrial context. The design strengthened coherence and professional relevance while making visible a productive tension between developmental learning space and workplace expectations of reliability and trust. For CDIO practitioners, the case illustrates how competence-oriented assessment artifacts and forward-oriented reflection can reconcile formative learning processes with industry expectations of credibility and trust.

Two targeted refinements follow from iterative delivery. First, to minimize cognitive overhead and keep thematic cycles in focus, mastery of collaboration tooling (e.g. GitHub) is moved to a prerequisite undergraduate course. Second, to accelerate effective collaboration and credibility, the opening session prioritizes team formation, trust-building, and exploratory brainstorming within the industrial context before readiness tests and application work.

Future work will evaluate how this sequencing affects team dynamics, credibility of deliverables, and competence development across cohorts and domains, and will explore how the assessment architecture scales to additional partners and related data-intensive courses.

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